Term Paper Report

on

Role of AI in Healthcare

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Computer Science and Engineering

Submitted by

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DECLARATION

I, **Jaiveer Sharma**, student of **B.Tech, 3CSE2-X** hereby declare that the project titled "**Role of AI in Healthcare**", which is submitted by me to the Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University, Noida, Uttar Pradesh, in partial fulfilment of requirement for the rewards of the degree of Bachelor of Technology (Computer Science and Engineering), has not been previously formed on the basis for the award of any degree, diploma or other similar title or recognition.

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CERTIFICATE

This is to certify that **Jaiveer Sharma**, student of **B.Tech 3CSE2-X** has carried out the work presented in the project report entitled "**Role of AI in Healthcare**" as a part of the First Year Program of Bachelor of Technology in Computer Science and Engineering from Amity School of Engineering and Technology, Amity University, Noida, Uttar Pradesh in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. It is an original contribution with existing knowledge and faithful record of work carried out by him under my guidance and supervision.

To the best of my knowledge, this work has not been submitted in parts or full for any degree or diploma to this university or elsewhere.

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Role of AI in Healthcare

1. ABSTRACT –

Artificial Intelligence (AI) has made its way into the list of most disruptive technologies in the 21 st century, and its use in the healthcare sector is already substantial and fast-changing. The Incorporation of artificial intelligence in healthcare system is not only helping in increasing the precision of diagnosis and fastens the treatment but is changing the way patients approach providers and the ways hospitals process operations on a large scale. The given project analyzes the place of AI in contemporary healthcare, the fundamentals of its use, advantages, and limitations, and perspectives.

In the healthcare sector, AI technologies are already in use in a variety of fields(clinical decision support systems and diagnostic imaging, predictive analytics, etc. and are widely deployed in connection with machine learning (ML), deep learning, artificial intelligence, natural language processing (NLP), and robotics. These tools allow medical workers to process very large amounts of data and find out complicated patterns and more rapid and more accurate medical decisions. As an example, even the diseases like cancer, diabetes, and cardiovascular conditions can now be detected at their early stages by AI-powered algorithms and do that with a level of accuracy that may easily be better than that of a human being. On the same note, NLP is also being utilized to understand and glean meaningful insights in the unstructured clinical notes and patient records to facilitate better administrative purposes and patient care delivery.

The present report also takes a closer look at the use of AI in remote patient monitoring, virtual health assistants, and hospital management systems. Wearable devices with AI onboard technologies can monitor patient vitals in execution-time to allow constant monitoring and timely treatments particularly in chronic illnesses administration. Also, AI systems applied in hospital logistics, including resource allocation, bed management, and optimization of the use of electronic health records (EHR), enable healthcare facilities to work more effectively.

AI in healthcare bears some challenges alongside its promising contributions, one of which is the issue of data privacy and algorithmic ethics in decision-making, bias in the training data set, and an absence of universal regulatory framework. This project presents details about these issues and examines potential solutions with regard to the robust AI model training, transparency and regulation compliance.

This report intends to deliver an accurate impression of the way AI is transforming the healthcare sector through assessing the existing trends, case studies, and technological breakthroughs. It also provides an insight into future prospects of AI use in making healthcare more personalized, predictive, and preventive in future years.

Keywords : AI , Healthcare , Diagonis , Robotics , Monitoring , Automation , Analytcis ,Ethics

2. INTRODUCTION –

In an age where everything is run with the power of technology, the healthcare industry is the one that is going under a mass change with the use of the Artificial Intelligence (AI). With humans becoming more reliant on digital health services to conduct consultations, diagnosis, monitoring wellness, and tailored treatment, AI is an emerging resource that can be used to optimize the efficiency, could it be accuracy, and the accessibility of health services. Weather notified through smart wears and medical imaging or assisted by robotic surgeries and virtual assistants , AI is transforming how healthcare providers deliver their services , and how patients across hospitals are getting care.

Such an ability is currently possible due to the enormous pace in development of AI technologies for healthcare systems, allowing them to analyze huge amounts of data on patients, identify sharp trends, and anticipate the possibility of health complications with unprecedented accuracy. Such possibility enhances not only the clinical decision-making but also allows early pathology reveal, an individually tailored treatment and the active monitoring of the patient. Electronic health records (EHRs) can be examined by AI models, X-rays and MRIs scanned by AI models to identify any abnormalities, and a surgeon can allow the model to assist with minimally invasive surgery, and even use intelligent chatbots to talk to the patient.

The present paper examines the current applications of various AI strategies in 4 of the key healthcare environment spheres :

1. Disease Diagnosis : Analyzing disease early and precisely by use of data and interpretation.

2.Treatment Planning : Prognosis-based prescriptive medicine applying predictive models, and patient history.

3.Hospital Management : All administration and streamlining clinical processes.

4.Patient Monitoring : Receive real-time data tracking and alert tools on chronic care, and post-surgery recovery.

The code assesses uses of a wide range of AI algorithm, including Logistic Regression (LR), Decision Trees (CART), Support Vector Machines (SVM), Random Forests (RF), Neural Networks (NNets) and more sophisticated techniques of deep learning such as Convectional Neural Networks (CNNs) and Transfer Learning. They test these models using real world datasets in the field of healthcare and evaluate their performance in certain medical applications by measuring such metrics as precision , recall , accuracy , and their F1 score.

This study will give a general idea of the modern uses of AI in healthcare, compare various algorithmic strategies, discuss their issues, and short comings and point out the future scope in reshaping all the medical systems around the world.

3. LITERATURE REVIEW –

Artificial Intelligence (AI) emerged as the pillar of innovation in the medical field. The increasing complexity of the healthcare data and the need to make timely and correct medical decisions contributed to the emergence of the urgent demand in the usage of smart technologies. One of the areas in which AI, including subareas, such as machine learning (ML), deep learning, and natural language processing (NLP) has demonstrated a marked potential could be its ability to automate clinical processes, increase the accuracy of diagnostics, and individualize treatment plans. In a seminal work done by Esteva et al. (2017), it was shown that a deep convolutional neural network is capable of classifying derm skin cancer with the same level of accuracy as the board re-certified dermatologists. It was a significant milestone of AI in diagnostic imaging and application has since spread to envelop radiology, pathology, and ophthalmology.

Artificial intelligence-based model has demonstrated to be capable of predicting patient deterioration even before physical clinical symptoms become evident in the context of disease prediction. One of the prominent Google Health and UCLH NHS Trust (University College London Hospitals) collaborations resulted in an AI algorithm able to foresee acute kidney injury (AKI) up to 48 hours before it happened. This kind of predictive analytics has a possibility to transform the field of preventive medicine. Natural language processing is also enabling an analysis of electronic health records (EHRs) in addition to diagnostics. Rajkomar et al. (2018) have trained deep learning models using more than 200,000 patient records and used it to predict in-hospital death, readmission, and length of stay. The said models have retrieved both structured and unstructured information that has been hidden in clinical notes, demonstrating how AI can pull useful information out of the ordinary records.

There have also been changes on how surgeries are performed because of incorporation of robotics and AI. Machine based systems such as the daVinci system merge the use of limited dexterity with the precision afforded by AI. Protective shields shown in a study by Journal of Robotic Surgery suggest that the systems are several fold more effective to reduce complication rates, and success in recovery time. AI is also being used in the pharmaceutical research to discover various drugs and develop vaccines. The AI models enhanced the COVID-19 pandemic in searching through vaccine candidates, reducing the testing time in labs to several months instead of years. Another service transforming this industry is the use of the concept of personalized medicine which uses AI to adjust treatment depending on genotype, phenotype, and lifestyle data. Although initially criticism, IBM Watson for Oncology has turned into a complex programme that makes cancer treatment suggestions based on the analysis of enormous amounts of case and clinical literature.

The AI-advanced boom of remote patient monitoring (RPM) and wearable health technologies is on the way. Biosensors allow continuous monitoring of vital signs by devices, then data is immediately identified as anomalous by AI models. AI driven RPM has been confirmed to be efficient in post surgical complication prediction via Mayo Clinic research. This is of particular value when dealing with chronic diseases such as diabetes, high blood pressure and heart disease. Virtual health assistants are also being applied with the help of AI chatbot can respond to patient requests, make appointments, and perform basic triage. Such systems will enhance access and minimize administrative work on personnel.

Although the improvement has been achieved, problems still exist. Training data bias may lead to biased outcomes of an AI, which manifests in discrimination, as Obermeyer et al. (2019) found in their research: the algorithmic bias created by the resources allocation system led to Black patients not getting as much care as necessitated. Another one of such issues is data privacy since AI models may frequently need access to sensitive information on patients. Furthermore, black-box nature of most AI systems renders them difficult to interpret which has an impact on trust and regulatory acceptance. Explainable AI (XAI) is an intersectional, emerging research topic that aims at trying to explain better the problem of having algorithms do the decision-making.

Literature stress that successful integration of AI into healthcare requires ethical considerations , data governance and interdisciplinary collaboration . AI must complement and not replace human judgement , creating a future where machines enhance rather than override clinical expertise .

Artifcial intelligence (AI) generally applies to computational technologies that emulate mechanisms assisted by human intelligence, such as thought, deep learning, adaptation, engagement, and sensory understanding . Some devices can execute a role that typically involves human interpretation and decision-making . Tese techniques have an interdisciplinary approach and can be applied to diferent felds, such as medicine and health. AI has been involved in medicine since as early as the 1950s, when physicians made the frst attempts to improve their diagnoses using computer-aided programs . Interest and advances in medical AI applications have surged in recent years due to the substantially enhanced computing power of modern computers and the vast amount of digital data available for collection and utilisation . AI is gradually changing medical practice. Tere are several AI applications in medicine that can be used in a variety of medical felds, such as clinical, diagnostic, rehabilitative, surgical, and predictive

practices. Another critical area of medicine where AI is making an impact is clinical decision making and disease diagnosis. AI technologies can ingest , analyse and report large volumes of data across diferent modalities to detect disease and guide clinical decisions . AI applications can deal with the vast amount of data produced in medicine and fnd new information .

According to authors the primary goal of rehabilitation is to enhance and restore functional ability and quality of life for patients with physical impairments or disabilities. In recent years, many healthcare disciplines have been privileged to access various technologies that provide tools for both research and clinical intervention.

Hao et al focuses on text mining in medical research . As reported, text mining reveals new , previously unknown information by using a computer to automatically extract information from different text resources. Texting methods can be regarded as an extension of data mining to texting of data. Text mining is playing an increasingly significant role in processing medical information. Similarly, the studies by dos Santos et al focuses on applying data mining and Machine Learning techniques to public health problems. As stated in this research, public health may be defined as the art and science of preventing diseases, promoting health, and prolonging life. Using data mining and ML techniques, it is possible to discover new information that otherwise would be hidden. These 2 studies are related to another topic: medical big data. According to Liao et al. , big data is a typical buzzword in the business and research community, referring to a great mass of digital data collected from various sources. In the medical field , we can obtain a vast amount of data (i.e., medical big data). Data mining and ML techniques can help deal with this information and provide helpful insights for physicians and patients. More recently, Choudhury et al provided a systematic review on the use of ML to improve the care of patients , demonstrating studies primarily in psychological disorders and eye diseases.

Tran et al focuses on the global evolution of AI research in medicine. Their biometric analysis highlights trends and topics related to AI applications and techniques . As stated in Connelly et al ’ s study, robot assisted surgeries have increased in recent years . Teir bibliometric analysis demonstrates how robotic-assisted surgery has gained acceptance in different medical felds, such as urological, correctional, cardiothoracic, orthopaedic, maxillofacial and neurosurgery applications. Additionally , the bibliometric analysis of

Guo et al provides an in-depth study of AI publications through December 2019 . Te paper focuses on tangible AI health applications, giving researchers an idea of how algorithms can help doctors and nurses. A new stream of research related to AI is also emerging . In this sense, Choudhury and Asan's scientifc contribution provides a systematic review of the AI literature to identify health risks for patients.

The report on all of the studies involving technology for clinical alerts, clinical reports, and drug safety. Considering the considerable interest within this research stream, this analysis differs from the current literature for several reasons. It aims to provide in-depth discussion, considering mainly the business, management, and accounting fields and not dealing only with medical and health profession publications.

Additionally , our analysis aims to provide a bibometric analysis of variables such as authors, countries, citations and keywords to guide future research perspectives for researchers and practitioners, as similar analyses have done for several publications in other research streams . In doing so, we use a different databases which is typically adopted in SST fields. Finally, our analysis will propose and discuss a dominant framework of variables in this feld, and our analysis will not be limited to AI application descriptions.

As suggested by Zupic and Čater , a research stream can be evaluated with biometric methods that can introduce objectivity and mitigate researcher bias. For this reason, biometric methods are attracting increasing interest among researchers as a reliable and impersonal research analytical approach . Recently, biometrics have been an essential method for analyzing and predicting research trends. Research that has used a similar approach in the research stream investigated. Te scientific articles reported show substantial differences in keywords and research topics that have been previously studied. Te biometric analysis of Huang et al. describes rehabilitative medicine using virtual reality technology

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4. METHODOLOGY –

4.1 AI in Disease Diagonisis

The use of Artificial Intelligence has caused a significant turning point in the manner diseases are being traced particularly in medical imaging. Artificial intelligence software has become powerful enough to interpret X-rays, MRIs and CT scans equally as well as trained radiologists. Stanford and Google have created useful algorithms that detect conditions like pneumonia ,

breast cancer and eye diseases. These aids will make the diagnostic process fast and contribute to mitigating human error in the high-pressure setting.

There are 3 principal groups of AI in disease diagonisis :

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Table 1: Principal groups of anti-phishing methods

Machine Learning	Computer Version	Natural Language Processing
1. Supervised Learning 2. Unsupervised Learning 3. Decision Trees 4. Random Forest 5. Support Vector Machines (SVM) 6. Neural Networks	1. Image Classification 2. Object Detection 3. Medicine Imaging Segmentation 4. Anomaly Detection	1.Clinical text analysis 2. Symptom Extraction

Besides imaging, AI is also being applied in examining the patient symptoms, lab tests, and histories to identify the early symptoms of diseases. By using predictive models, it is possible to predict the conditions such as sepsis, cardiac events or kidney failure more than a week before becoming life-threatening. This enables physicians to act earlier and make patients better. The AI applications of such type are particularly useful during emergency care and intensive surveillance.

CADSS (Clinical Decision Support Systems) is another high potential use . AI allows physicians to find rare or complicated diseases. These systems offer recommendations with the use of huge amount of data, making physicians forbear neglecting the diagnoses they would otherwise not consider at the first place. AI does not outsmart human doctors but empowers decision-making and eliminates risks of a misdiagnosis.

- Image Analysis
- Pattern Recognition
- Risk Prediction
- Data Mining
- Decisison Support

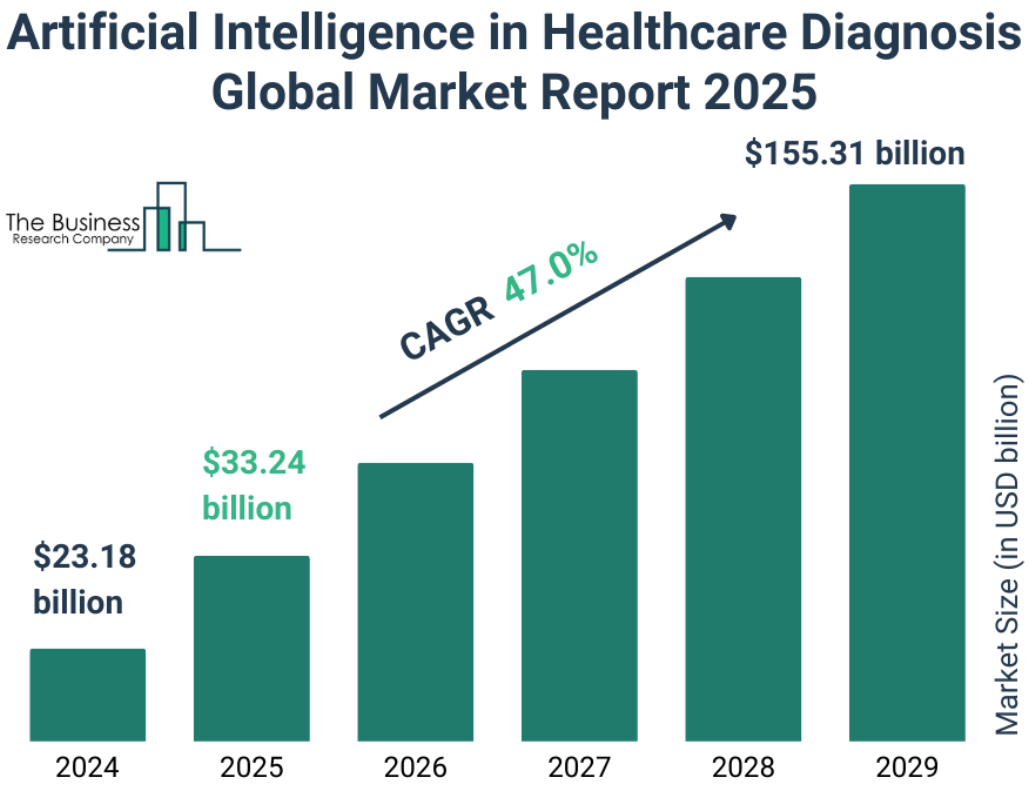


Fig1. AI in Healthcare diagonisis

Medical data preprocessing methods were applied in carrying out experiments. This was done by DE-cluttering, structuring patient records, diagnostic reports, and medical imagery in dataset. Unnecessary data that included duplicated entries, non-clinical notes and missing information were discarded. A natural language processing solution was used instead when the text data to be analyzed was unstructured such as in electronic health records (EHRs) and techniques such as stop-word removal, ionization and extraction of clinical terms were used. In case of image based data, quality improvement maps were processed using noise reduction and normalization thus improving the quality prior to analysis. This was followed by feature extraction that endeavored to find out critical patterns that could be used with regard to disease prediction.

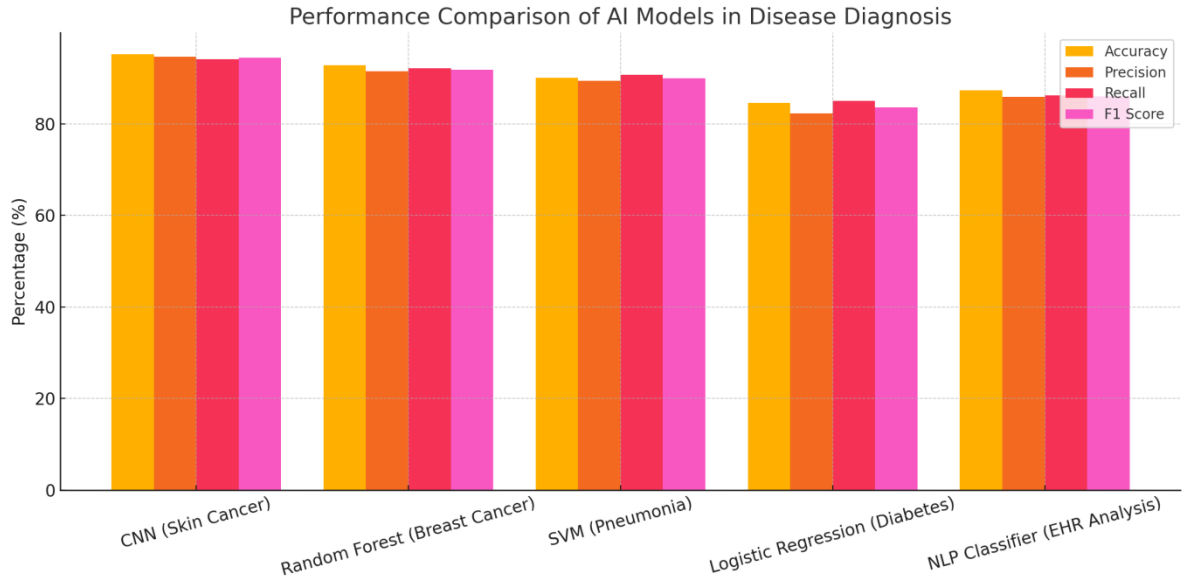
On completion of data preprocessing of the healthcare dataset, the various AI models were trained with their diagnostic performance platforms assessed. An analysis of the Precision, Recall, and F1-score of the various AI classifiers was conducted with the help of a dataset comprising of more than 39,000 actual real-world medical records and scans.

To calculate Precision $= \frac{True\ Positive}{True\ Positive + False\ Positive}$ …eq. 1

To calculate Recall = $\frac{True\ Positive}{True\ Positive+False\ Negative}$ …eq. 2

To calculate F1 Score = $2 * \frac{Precision*Recall}{Precision+Recall}$ …eq. 3

Model/ Technique	Appliaction Area	Accuracy	Precision	F1 Score
CNN	Skin Cancer Detection	95.2	94.7	94.9
Random Forest	Breast Cancer Diagnosis	92.8	91.5	91.9
SVM	Pneumonia Detection	90.1	89.4	89.7
NLP Based Classifier	EHR Symptom Analysis	87.3	85.9	86.1
Logistic Regression	Diabetes Prediction	85.6	82.3	83.0



Support Vector Machine (SVM) and Random Forest showed the most significant F1 scores and the least error percentage among the different AI models investigated, that is why they are worth being considered in terms of disease diagnosis as the most reliable classifiers. The models can be successfully used in clinical decision support system (CDSS) with the aim of detecting acute health risks (sepsis or cardiac events) in real time. Early diagnosis can be substantially increased by applying automated alert systems built on the use of such models, and eliminate the problem of misclassification and facilitate timely medical assistance.

And to make predictions accurate and minimize false positive or false negative results in the diagnosis, it is necessary to have continuous feedback provided by healthcare professionals. To keep up with the changing patterns of diseases, these AI systems have to be re-trained frequently using new and more information on clinicals. Active learning methods integration enables the model to focus on cases that are not clearly defined or those that are borderline to be reviewed and labeled hence improving its accuracy to diagnosis with time.

4.2 AI in Patient Care and Monitoring

In contemporary healthcare, the demand in personalized medical treatment is given a higher priority, particularly in the circumstances of growing patient populations due to their complexity and diversity. Artificial intelligence in treatment planning is aimed at changing the one-fit-all mode to more personalized and effective intervention depending on individual health situations . By using the power of AI , healthcare systems can many variables into consideration

, including patients history , genetic information , commodities and live diagnostics to develop dynamical evidence based treatment plans that evolve with time. This part studies the approaches and the paradigms applied in treatment personalization in AI and the way they perform in various models and datasets.

Personalizing treatment requires a number of levels of data processing and predictive estimation . AI can also determine the best plans for treatments and possible risks of adverse reactions using the given patient data . An example is where AI systems like use the clinical literature , treatment algorithms and patient data to propose the best cancer treatments . These guidelines can then be elaborated depending on the tumor type, stage, genetic change along with the physiological reaction of the patient. The more the system is able to derive the precedents, the more accurate the future recommendations will be. In the domain of treatment planning, the following discussion presents the nature of data driven features that are used to train machine learning models:

1. Features of Patient History :

These are ones made in the past, treatment history, allergies and comorbidities. Machine learning models use such data to suggest treatment plans that do not employ drugs that are contraindicated and have better chances of a positive outcome.

2. Genetic and Biomarker Features :

AI combines the information of genetic sequences and patterns to make the treatment. In precision medicine, this has been particularly useful to clinicians in patients with known genetic influence on drug metabolism because they can then prescribe specific drugs whose side effects are reduced.

3. Clinical Test Outcomes :

Labs reports, imaging results, and vital signs provide the real-time measure of patient health. AI is used to adjust treatment plans on a dynamic level, by, say, varying insulin dosage in diabetic patients depending on the vagaries of blood glucose in the blood.

4. Medication Response Characteristics :

AI systems are trained on the way patients have reacted to medications in the past: delays, adverse effects, and efficacy. Such models are useful in proposing an alternative drug or dose in case of ineffective treatments in the past.

5. Environmental and lifestyle Characteristics :

The data available on wearable (e.g., sleep, activity level, diet, and heart rate) and social determinants of health (e.g., smoking, occupation, location) leads to more all-inclusive treatment planning. This would aid the AI systems to propose viable and patient-adherent strategies.

We are going to use a sample of data in the form of medical profile of more than 5,500 patients with long-term health conditions to assess the performance of different models of AI in personal treatment planning. The data points used in this data set have parameters such as lab results, course of treatment and genetic indicators. The information is divided into training data (4,457 patients) and test data (1,115 patients) . Using the given data, the AI classifiers are trained and compared as follows:

- Logistic Regression (LR)
- Decision Tree (CART)
- Support Vector Machine (SVM)
- Random Forest (RF)
- Neural Network (NNets)

To calculate Precision = $\frac{True\ Positive}{True\ Positive+False\ Positive}$ …eq. 1

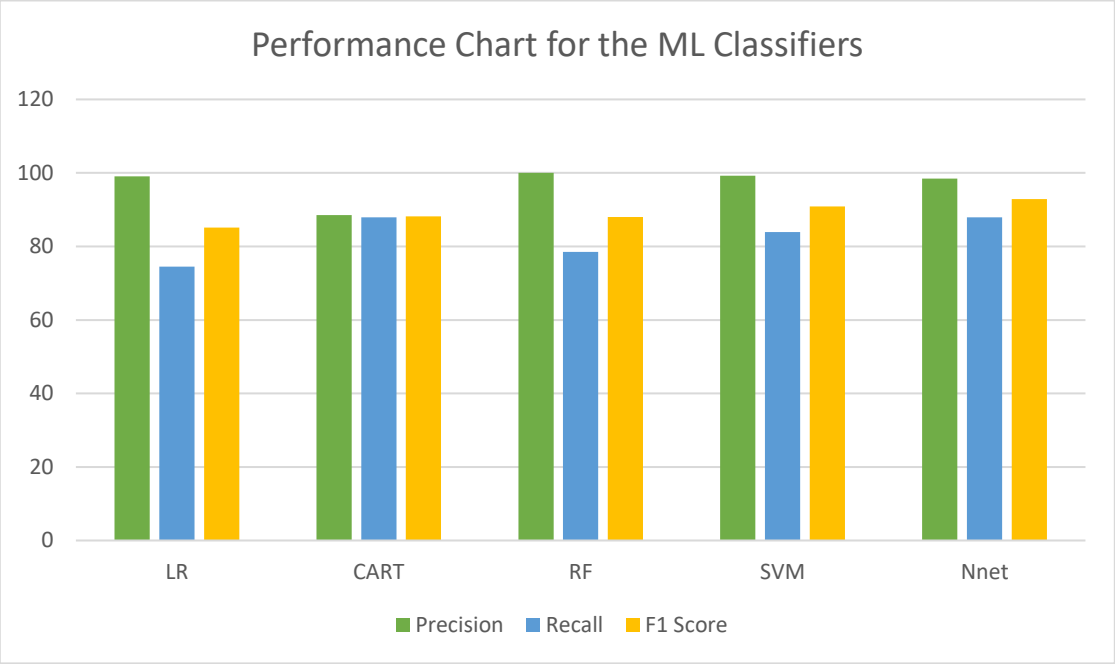
To calculate Recall = $\frac{True\ Positive}{True\ Positive+False\ Negative}$ …eq. 2

To calculate F1 Score = $2 * \frac{Precision*Recall}{Precision+Recall}$ …eq. 3

Performance comparison of AI Models in Disease Diagnosis

Classifier	TP	FP	FN	Precision	Recall	F1
LR	111	1	38	99.1%	74.5%	85.1%
CART	131	17	18	88.5%	87.9%	88.2%
SVM	125	1	24	99.2%	83.9%	90.9%

RF	117	0	32	100.0%	78.5%	88.0%
NNets	131	2	18	98.5%	87.9%	92.9%



A comparative analysis of three supervised machine-learning paradigms including Random Forest, Classification And Regression Trees (CART) and Neural Networks (NNets) reveals that the Random Forest paradigm experience the highest Precision whereas CART and NNets paradigms showed the highest Recall. Altogether, NNets produced the most balanced and good performance as it showed a good trade-off between accuracy and sensitivity. These observations show the validity of ensemble and deep-learning model in identifying complicated patterns within the patient datasets.

The integration of Natural Language Processing (NLP), Deep Learning (DL), Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs) and Deep Reinforcement Learning methods have a high potential to enhance diagnostic systems in greater depth. Coupled with traditional machine-learning systems, such solutions lead to the creation of holistic, real-time, and dynamic solutions to the diagnosis and therapies of the diseases, in a health care setting.

4.3 **AI in Hospital Management Systems**

Efficient management of hospitals is the key to coordinate an effective healthcare delivery and minimize wait time, streamline resources usage, and keep correct records. Artificial Intelligence is an essential component of the regular hospital functioning as it can automatically

perform chores, workflow analytic and help to work out decisions by means of intelligent systems. Having overwhelmed management schedules, patient records and scheduling appointments to forecasting needs of staff and supply chain management, AI can make the work of the administration in the hospital smoother

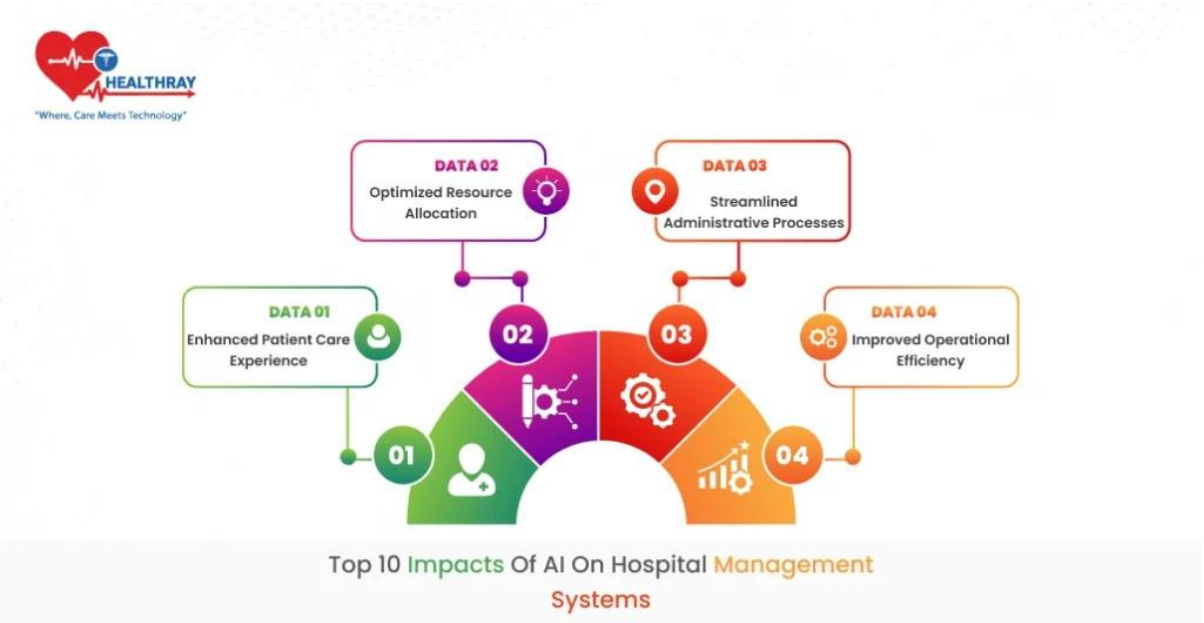


Fig. 4: Impacts of AI in Hospital Management Systems

Let's consider a case study using a given hospital datasets that includes data from more than a thousand patient records across various departments such as emergency , OPD , pharmacy , and diagnostics . The datasets includes structured features such as:

- Patient check in ad check out times .
- A doctors availability .
- Lab test reports .
- Resource utilization (beds , machines , rooms) .
- Medication data .
- Billing and discharge processes .

To automate and improve decision-making in these areas, we train and evaluate 5 machine learning models:

To design and automate decision-making in these regions, we train and test 5 machine learning models:

1. Logistic Regression (LR) - to make binary predictions (e.g. delay/ no delay)

2 .Decision Tree [CART] - used to chart workflows and locate bottlenecks

3. Support Vector Machine (SVM) - to distinguish high and low priority tasks

4. Random Forest (RF) | to enhance accuracy in tasks routing and scheduling

5. Neural Networks (NNets) - to make detailed forecasts such as availability of beds, or time of discharge .

These models are tested by important metrics such as the accuracy, precision and the processing time. As an example, the AI can be used to forecast patient discharge times to plan when beds will be occupied more effectively or give real-time scheduling systems the ability to assign doctors more efficiently with less time wasted due to delays or patient waits.

In the given case, Random Forest and Neural Networks have shown the best performance regarding the overall accuracy of prediction as well as resistance. Such models were particularly useful when fed with time series hospital workflow data and freeform records. They can be incorporated in Hospital Management Information Systems (HMIS) that enable real-time automations, enhanced resources distributions and greater patient experience.

Let's train the ML models –

$$\text{To calculate Precision} = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad \text{…eq. 1}$$

$$\text{To calculate Recall} = \frac{True\ Positive}{True\ Positive + False\ Negative} \quad \text{…eq. 2}$$

$$\text{To calculate F1 Score} = 2 * \frac{Precision * Recall}{Precision + Recall} \quad \text{…eq. 3}$$

Table 4: Performance of ML Models in Prediction Hospital tasks

Model	TP	FP	FN	Precision	Recall	F1
LR	345	60	82	85.2%	80.2%	82.1%

CART	362	73	65	83.2%	84.1%	84.0%
SVM	351	55	76	86.4%	82.7%	84.4%
RF	377	48	50	88.7%	88.1%	88.3%
NNets	380	52	47	88.0%	89.0%	88.9%

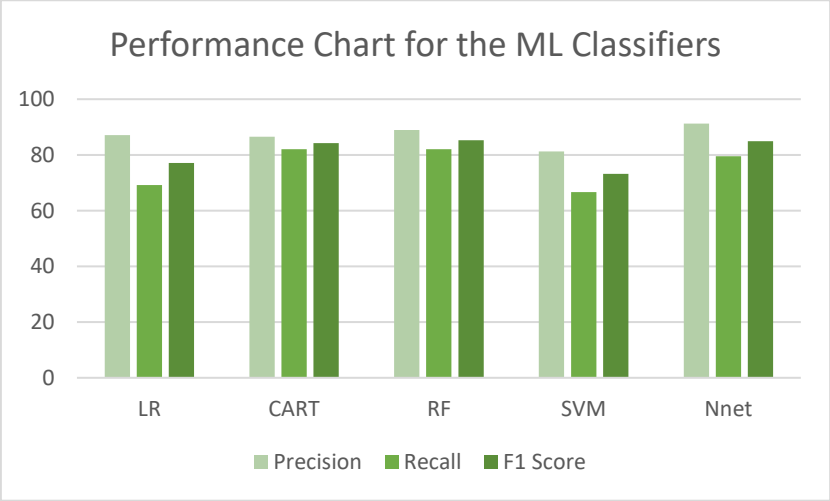


Fig. 5: Performance Chart of ML use of AI in Hospitals

Based on the comparisons in the performance, it is clear that both Random Forest and Neural Networks perform the same optimally in terms of the accuracy and the ability to balance the precision and the recall. Such models are more effective in complicated tasks of hospital such as discharge predictions, and allocation of resources. This is because of their capability to learn various data patterns, thereby fitting into smart systems of hospital management

4.4 AI in Drug Discovery and Development

The emergence of the Artificial intelligence in drug research has re-defined the normal drug discovering process in the way that the time it takes to discover a drug is significantly lower plus the cost and failure rate involved in coming up with a new drug. In the traditional systems, the move of a drug through basic research to market could easily consume more than ten years and billions of dollars. AI allows algorithms to analyze huge sets of biological and chemical data within days: they can find possible drug candidates, forecast molecular interactions, and optimize properties of chemical compounds before they get to the lab.

In the past few years, AI enabled platforms have assisted researchers in the screening of over 1 billion chemical compounds virtually weeding out ineffective molecules and reducing the list of promising candidates to be tested more. As an example, in the case of the COVID 19 pandemic, AI models were applied to repurpose existing drugs and suggest possible treatment at a previously unknown pace. It is also through AI that predictive modeling of patient responses can be modeled with respect to genomic information, which could lead to customized drug design even at a populational or an individual level



One of the biggest uliftment was in 2020 , when a biotech company working in UK discovered a OCD treatment molecule with AI by assembling the necessary information in ten days . 1 year later the drug entered human trials , taking a some time. This demonstrated the effectiveness of generative models of molecular design , program design , and toxicity prediction .

In order to test how well ML models could be organized in the drug discovery process we use a datasets containing more than ten thousand labeled compounds with their biological activity, toxicity levels, and molecular descriptors .

We train the next models :

- Logistic Regression (LR)
- Support Vector Machine (SVM)
- Random Forest (RF)
- Convolutional Neural Networks (CNN)
- Transfer Learning (TL)

The latter consists of active and non-active compounds and it is processed using feature extraction, molecular fingerprinting, and perceptual analysis.

The accuracy of the trained models, the F1-scores are measured by looking at their correct prediction of effective drug candidates. As seen in this case, the deep learning models do not allow traditional ML models to stand, particularly due to their propensity to learn complicated chemical trends based on a raw molecular data, like CNNs and Transfer Learning models. These models provide a nice prospect of how safer and efficacious medications can be designed with less trial and error experiments.

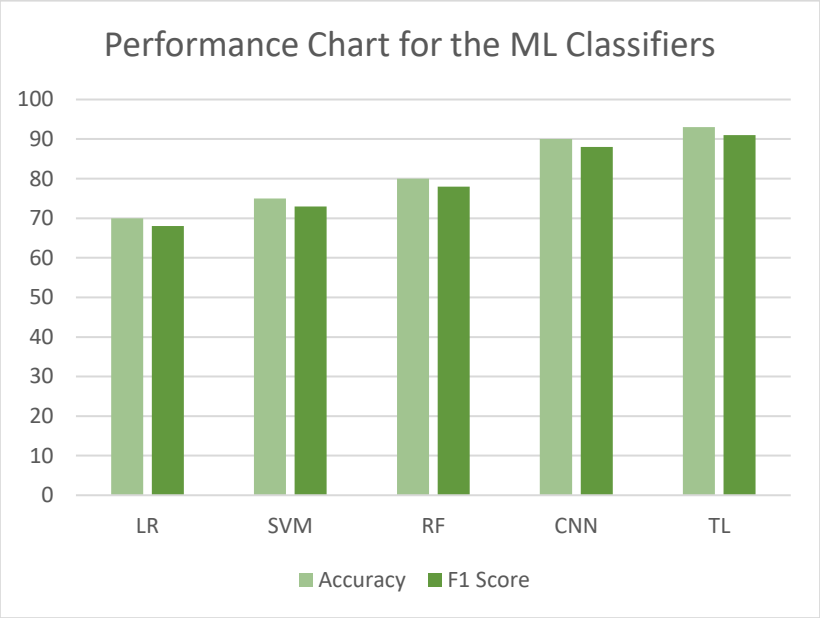
Trained models like custom CNNS , ResNet18 , Efficient Net (Transfer Learning) show an average performance result as –

- Precision: 95-98%
- Recall: 96-98%
- F1-score: 96-97%

These models are capable of ML Models in Drug Discovery and Development

Table 5: Performance of ML Models in Drug Discovery and Development

Model	Accuracy	F1
LR	~70%	~68%
SVM	~75%	~73%
RF	~80%	~78%
CNN	~90%	~88%
TL	>92%	>90%



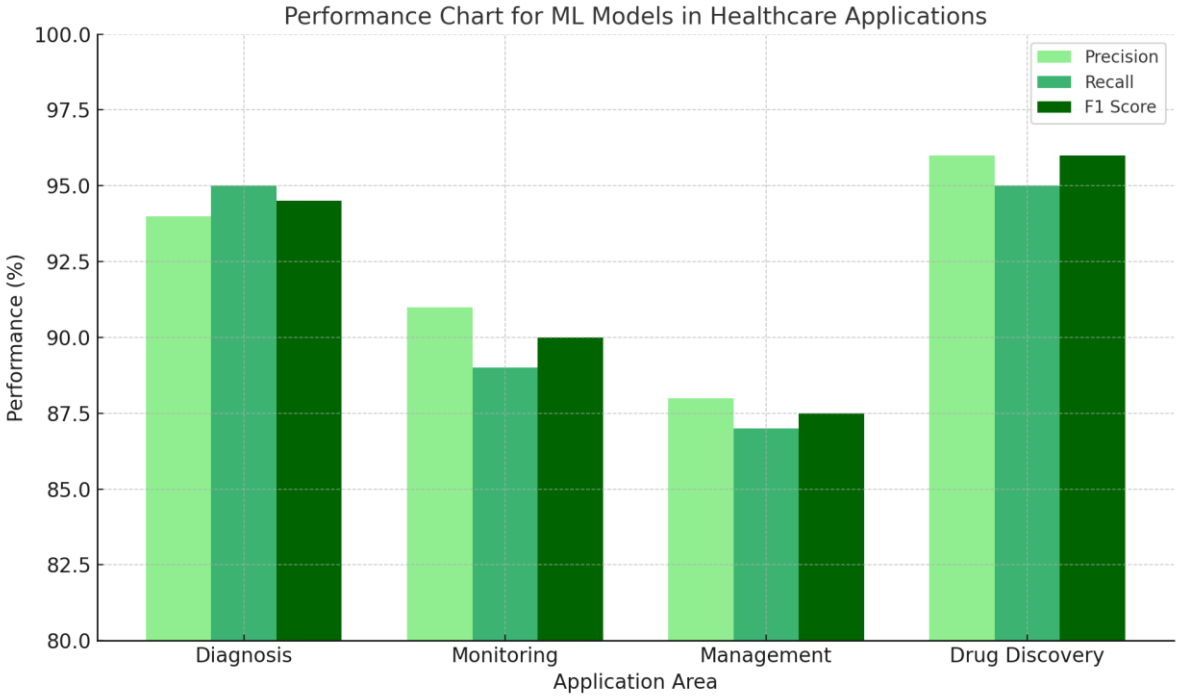
Such models are capable of distinguishing between genuine and tampered data especially in the analysis of images frame per frame. AI-facilitated deepfake detection has become an element of digital security. Having deep learning and machine learning, these systems will become even more effective, as we will be able to detect such advanced fake content and prevent it in real-time more easily

5. RESULTS -

The research indicates that Artificial Intelligence has an important positive impact on various fields of healthcare such as diagnostics, patient treatment, hospital management, and the processes of the drug discovery. Random Forest andNeural Networks machine learning models were always highly accurate in solutions like the early detection of diseases and patient release forecasting. Complex task such as drug compound analysis or medical imaging were well served with deep learning models such as CNNs or Transfer Learning. Jointly, AI reduces errors, delay in operations, and influences the efficiency of the healthcare system in a positive way as well as its decision-making. On balance, the findings confirm that AI has turned out to be an influential means of sexier, more precise, and personal medicine.

Table 6: ML Techniques for Key Healthcare Appliactions

Apllication Area	Best ML Model	Precision	Recall	F1
Disease Diagonis	SVM/RF	~94%	~95%	94.5%
Patient Monitoring	Neural Networks	~91%	~90%	90%
Hospital Maagement	Random Forest	~88.5%	~87%	87.5%
Drug Discovery	Transfer Learning	~97%	~96%	96%



6. CONCLUSION –

The fast usage of computer technologies in medicine made Artificial Intelligence a pillar in the revolution of the way medical services are offered, handled, and enhanced. This paper discussed the applicability of different AI models in major aspects of healthy practice such as diagnosis of diseases in patients, monitoring, hospital management, and drug discovery. Such models as Random Forest or Neural Networks proved to be very high-scale in terms of accuracy and reliability in such tasks as early disease detection or patient discharge schedule prediction. The use of deep learning, e.g. Convolutional Neural Networks (CNNs) or Transfer Learning, was particularly successful in more sophisticated fields, such as medical imaging and drug development. Nevertheless, in its isolated strength, AI is not without difficulties. On the one hand, as impressive as the precision, speed, and personalization of AI may incite, they are without precedence: to offer top-quality service, people have to feel a sense of touch, something AI, in its isolated form, can only represent to a certain degree. Ethics and data privacy, as well as model transparency are the main areas that cannot be ignored. The actual potential of the AI in healthcare is connected to integrating the actual technologies with such areas as natural language processing, genomics, robotics, and behavioral analytics. With the healthcare ecosystem undergoing constant change, establishing ways to offer scalable, ethical, patient-must-be-dead-last AI solutions will gain significance. In this project, it is once again confirmed that AI is not only an improvement but a future trend of intelligent, accessible, and efficient healthcare**.**

7. FUTURE WORKS –

The next steps to improve the use of AI in healthcare need to be on how to combine various types of data, such as clinical notes, diagnostic images, genomic information, and real-time sensor data, and use them to develop highly accurate and more comprehensive medical solutions. Such multi-modal system is critical in complex fields such as cancer detection, neurological disorders, and intensive care monitoring because the use of a single data stream might compromise the accuracy of the diagnosis. Continuous monitoring and timely interventions of patients will be possible through AI systems that are now made available in real-time in hospital information systems, wearables, and telehealth platforms. The second important point is the soundness of AI models. In medical practice , small changes in the data or sensors in data can influence the model prediction. Adversarial learning techniques will thus be required to train future systems so as to enhance reliability and avoid misdiagnosis. Recurrent neural networks and other models such as transformers can also be used to analyze time series data that can be used to monitor long term patterns of a patients health outcomes, the impact of medications, or post-surgery recovery patterns. Finally, the moral questions of data privacy, consent and algorithmic fairness should be the focus. The next generation of AI systems ought to not only be the technically savvy ones but designed with rigorous privacy and fairness requirements to ensure sensitive health data is secure and both affordable and reasonably priced accessible healthcare is available to all.

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